



School of Computer Science and Engineering (SCOPE)

BACSE291 – Innovative Design Project

B.Tech. (CSE & Specialization)

Academic Year 2026–2027

Innovative Design Project — Student Guidelines

Understanding the Review Structure:

- The project is evaluated across **5 stages** totaling **100 marks** — treat every review as a mini-viva, not just a submission deadline.
- **Zeroth Review (5 marks)** — Guide only; submit a crisp title and a well-written abstract (problem, approach, expected outcome in ≤ 250 words); due **on or before 7 September 2026**.
- **Review 1 (20 marks)** — Panel review on **7 & 14 October 2026**; focus is entirely on domain knowledge, literature survey, research gaps and objectives — code is NOT expected here.
- **Review 2 (30 marks)** — Panel review in **January 2027 (2nd/3rd week)**; at least **60% of your proposed methodology must be implemented** with a complete system design.
- **Report Preparation (5 marks)** — Submit full report and rough outcome draft **on or before 15 March 2027**; guide verifies and approves.
- **Final Review (40 marks)** — Examiner panel in the **last week of March 2027**; complete presentation with live code demo and proof of outcome — this alone carries 40% of your grade.

Zeroth Review — What to Prepare

- Choose a **genuinely novel problem**, not a tutorial replication
- **Title must be specific**: avoid vague titles like "A Study on AI for Healthcare" — write "Federated Learning-Based Arrhythmia Detection on MIT-BIH with On-Chip Inference" style titles
- **Abstract must cover**: background → problem gap → your approach → expected measurable outcome

Review 1 — Literature & Gap Analysis

- Survey a **minimum of 25 peer-reviewed papers** (SCI/Scopus-indexed journals preferred; reputed conference papers admissible)
- At least **70% of papers must be from the last five years**; foundational/seminal works can be older but must be explicitly mentioned
- Do not write a narrative list of summaries — build a **comparison table** with columns: Author & Year / Method / Dataset or Testbed / Key Parameters / Limitation
- **Research gaps must be derived directly from the comparison table** — each gap should point to specific papers
- Frame **3–4 specific, measurable objectives** — objectives should state outcomes, not activities (wrong: "to study deep learning"; right: "to achieve $\geq 90\%$ F1 on the X dataset with an inference latency below 10 ms")
- Prepare to answer domain-knowledge questions from the panel — every team member must understand the full project, not just their assigned module

Review 2 — Implementation Progress

- **60% implementation is the minimum bar** to pass this review
- Your presentation must cover all four of: Project Plan / Analysis & Requirements / System Design / Proposed Methodology
- Show **running code, partial results and at least one measured metric** — even a preliminary result is better
- Maintain a **GitHub repository** with commit history — the panel may check that work was done progressively
- Justify every design decision: why this algorithm, why this dataset, why this tool — panels penalize teams who cannot explain their own choices

Report & Final Review

- The report must follow the standard format. Template will be shared.
- Results must include **tables and graphs** with labelled axes, units, legends and captions
- **Every result must be compared against a baseline**
- Proof of outcome for publication-track projects: acceptance email, submission ID, or a complete manuscript in the venue template
- Proof of outcome for patent-track projects: filed application number or a complete provisional specification draft
- **All team members must present and answer questions** during the final review.

General Do's and Don'ts

Do:

- Start the literature survey in Week 1 and study each paper
- Assign clear module ownership to each team member from Day 1
- Maintain a weekly log of work done and get verified from guide
- Pick an open dataset early so you are not blocked on data collection
- Aim for a publishable angle that is **narrow and new**, not broad and obvious

Don't:

- Copy code from GitHub and present it as your own implementation — panel members can identify this
- Leave integration of modules to the last week — interface bugs take as long to fix as the modules themselves
- Submit a report without proofreading — grammar and formatting errors in a research report reflect on your academic seriousness
- Rely on a single team member for all coding or all writing — interdependence is the point of a team project
- Change your project title or domain after the Zeroth Review without explicit guide approval